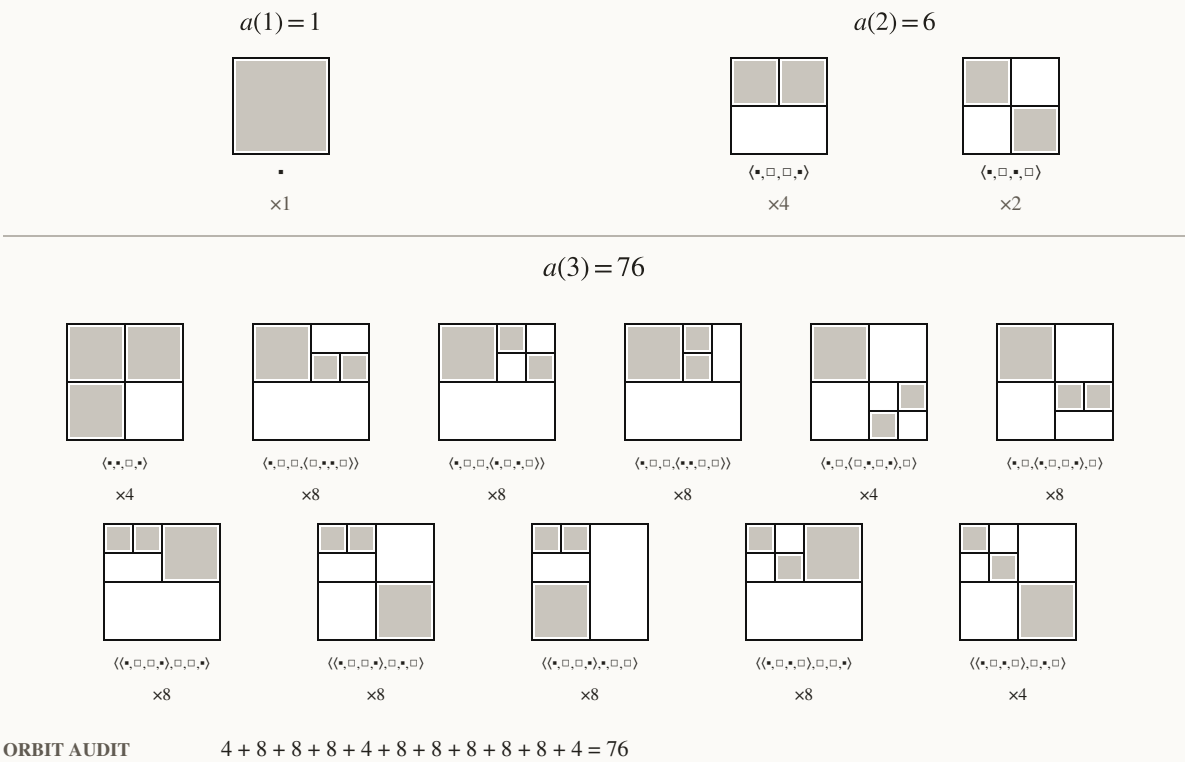


# Counting the quadrant trees

A combinatorial derivation



D<sub>4</sub> representatives; multipliers recover placements. Codes run CCW: NW → SW → SE → NE.

A quadtree square needs to be split into quadrants, possibly recursively, if it contains two or more points. Apply this procedure recursively to obtain a perfect quaternary tree pointing to  $n$  points on  $n$  true leafs. Forget those branches not ending at a true leaf point.

Now let  $i, j, k$  count nodes with 2, 3, or 4 valence only, and distinguish branch-like ( $b_x$ ) from true leafs ( $l_0$ ). Each branch introduces a differential toward leaf count:

$$b_x : X \longmapsto X' = X + x - 1$$

Summing over all branch nodes, starting from one root and ending in  $n$  true leafs we obtain that:

$$1 + i + 2j + 3k - n = 0$$

The sum is taken over all non-negative indices subject to the zero-sum constraint:

$$a(n) = \sum \frac{(n+i+j+k-1)!}{n! \, i! \, j! \, k!} 6^i 4^j, \quad \text{where } i, j, k \geq 0, \quad 1 + i + 2j + 3k - n = 0.$$

The total number of nodes including the root and true leafs will be  $m=i+j+k+n$ . The raw multinomial ordering of these symbols is:

$$\binom{m}{n, i, j, k} = \frac{m!}{n! \, i! \, j! \, k!}$$

For each sequencing of  $b_x$  and  $l_0$  symbols there is exactly one cyclic rotation whose partial sums are non-zero until all leafs are consumed, so we divide the multinomial by  $m$ .

The  $b_2$  and  $b_3$  branchings are associated to 6 and 4 distinct realizations in space, so we multiply by  $6^i$  and  $4^j$  to obtain the final closed-form summand:

$$\frac{(n+i+j+k-1)!}{n! \, i! \, j! \, k!} 6^i 4^j$$

# The witness certificate

Just the integral-differential facts

## INTEGRAL FORM

$$D(u)=1-6u-4u^2-u^3, \quad H_n(u)=\frac{1}{nu^nD(u)^n}, \quad q_n=\frac{1}{2\pi i}\oint H_n(u)du$$

Put  $B(u)=6u+4u^2+u^3$ , so  $D(u)=1-B(u)$ . Expanding  $(1-B)^{-n}$  and then  $B^m$  gives the Laurent series

$$q_n=\frac{1}{2\pi i}\oint \sum \sum \binom{n+m-1}{i,j,k}6^i4^j u^{i+2j+3k-n}du, \quad \text{where } m,i,j,k\geq 0, \ i+j+k=m.$$

A normalized contour integral around  $u=0$  keeps exactly the  $u^{-1}$  term:

$$\frac{1}{2\pi i}\oint u^rdu=1 \quad (r=-1), \quad 0 \quad (r\neq -1).$$

Thus the exponent condition moves underneath the sum:

$$i+2j+3k-n=-1 \iff 1+i+2j+3k-n=0,$$

$$q_n=\frac{1}{n}\sum \binom{n+m-1}{i,j,k}6^i4^j, \quad \text{where } m,i,j,k\geq 0, \ i+j+k=m, \ 1+i+2j+3k-n=0.$$

$$\frac{1}{n}\binom{n+m-1}{i,j,k}=\frac{(n+i+j+k-1)!}{n!i!j!k!} \implies q_n=a(n).$$

## EXACT VERIFICATION IDENTITY

$$\sum_{r=0}^3 P_r(n)H_{n+r}(u)=\frac{d}{du}(R(n,u)H_n(u))$$

## RATIONAL CERTIFICATE

$$R(n,u)=\frac{N(n,u)}{u^2(u^3+4u^2+6u-1)^2}$$

$$\begin{aligned} N(n,u) &= n^2(64u^9+576u^8+2304u^7+4496u^6 \\ &\quad +2784u^5-5136u^4-8524u^3+204u^2+6396u-491) \\ &+n(112u^9+1008u^8+4032u^7+7988u^6 \\ &\quad +5784u^5-6228u^4-11872u^3-948u^2+6648u-491) \\ &+40u^9+360u^8+1440u^7+2960u^6 \\ &\quad +2960u^5+640u^4-440u^3+40u^2. \end{aligned}$$

## ALGEBRAIC & DIFFERENTIAL OPERATORS (from OEIS A120593)

$$\begin{aligned} 5A(x) &= 4+x+A(x)^4 \iff Q=A-1, \quad Q=\frac{x}{D(Q)} \\ (256x^3+3072x^2+12288x-491)A^{(3)} &+ (1152x^2+9216x+18432)A^{(2)} \\ &+ (688x+2752)A^{(1)} - 40A = 0 \end{aligned}$$

## REFERENCE

H. Brochet and B. Salvy, “Reduction-Based Creative Telescoping for Definite Summation of D-finite Functions,”

<https://arxiv.org/abs/2307.07216>

## RECURRENCE OPERATOR

$$\sum_{r=0}^3 P_r(n)q_{n+r}=0$$

$q_0, \dots, q_5 = 1, 1, 6, 76, 1201, 21252$   
 $n=0$  is checked directly.

$$\begin{aligned} P_0 &= -8(4n+5)(2n+1)(4n-1) \\ P_1 &= -64(n+1)(48n^2+96n+43) \\ P_2 &= -6144(2n+3)(n+2)(n+1) \\ P_3 &= 491(n+3)(n+2)(n+1) \end{aligned}$$